Deliverable #1 – Technical Report

Project 13: Disaster Relief Donation Logistics | Client: Egyptian Relief Charities Coalition (ERCC) | Sponsor: Heba Elmasry

C-SW312 Introduction to Software Engineering, Fall 2026 | Team 12 | Egypt University of Informatics, Faculty of Computing and Information Sciences

1. System Vision Document

We're calling it ReliefLink for now, just a placeholder.

The problem? Each of the 14 charities keeps its own Excel sheets, so nobody ever sees the full picture. Last year, two charities sent blankets to the same camp, while another camp got nothing. One warehouse overflowed with items nobody asked for. Food expired because people sent out the newest boxes first. Some families got help twice, just because they were registered with more than one charity. And donors who wanted to know where their money went hardly got any real answers.

Our vision: one app that brings everyone together (charities, field teams, warehouses and donors). It connects what people need with what people donate, helps plan convoys that cross the border, and tracks every item from the donor straight to the family. It keeps working even when there's no internet. Donors get to see proof their donations arrived, without seeing who the families are.

How the system works

- Field coordinators fill in needs forms on their phones, offline if needed. The Planner approves each one, turning it into an open request that all charities can see.
- Donors can pledge goods or money. Goods must match an open request, and if someone pledges more than 1,000 items, they book a delivery slot at the warehouse.
- Warehouse staff check everything that comes in and refuse expired or damaged goods. Accepted goods get Arabic and English labels, and the system tells staff which batch should leave first.
- The Logistics Officer plans convoys with truck limits, border quotas and permits in mind, and can reschedule quickly if a permit gets turned down.
- At distribution points, handovers are recorded, even offline, and the system checks the 14-day rule across all charities.
- Money donations become purchase orders, but anything over 500,000 EGP needs Finance Committee approval. Donors get a PDF report in both Arabic and English.

We'll know the system works when

- No request is covered twice, and no family receives the same category of aid within 14 days.
- Every item can be traced from pledge to handover, and each donor gets a report about their own donation.
- Serving a family takes under a minute, and handovers still work offline all day.
- We never save photos of identifiable families without consent, and never any photos of children.

Scope and limits: We're aiming for a first version in four months, covering needs, pledges and warehouses. Convoys and distribution come later. There are 3 warehouses (Cairo, Ismailia, Arish) holding about 18,000 pallets, and a convoy can have around 60 trucks. Every charity has 2 to 10 staff users plus volunteers on phones, and distribution points have no reliable internet. The system must comply with the Beneficiary Data Protection Policy (ERCC PRT 01). We don't handle border procedures or driving the trucks.

2. Stakeholder Identification

We grouped the stakeholders into the four categories from Chapter 2.

Group	Stakeholders	What matters to them in the system
Internal operational	Field coordinators (like Mariam Sami), warehouse manager and staff (Ahmed Badawi), Logistics Officer, Coalition Planner, charity staff and volunteers	These people use it every day, so they want it fast and simple: the offline needs form, scanning to record a handover, easy receiving with Arabic labels, loading convoys, and syncing that works. Basically they want to stop running everything through Excel and WhatsApp.
Internal executive	Heba Elmasry (sponsor), Hesham Kandil (Finance Committee), Rasha Nour (Donor Relations), Dr. Tamer Hosni (Protection Officer), the Coalition board	They want one view of what is needed and what is already on the way, so work is not done twice. Approvals have to be written down. They also care about donor reports, dashboards, the four-month deadline and keeping family data safe.
External operational	Beneficiary families (including elderly and disabled people), donors (people and companies), truck drivers, local suppliers	Families want quick, respectful service and a backup if the QR card gets wet or torn. Donors want their delivery slot confirmed and to know where the donation ended up. Drivers and suppliers need clear cargo lists, permits and purchase orders.
External executive	Border authority, international organizations and big donors, aid coordination groups (like the UN cluster system), Egyptian authorities	The border authority needs correct cargo lists and wants quotas respected. Big donors want proof of delivery and honest handling of the money. Coordination groups and authorities care about no overlap between agencies, Sphere standards and legal handling of family data.

3. Functional Requirements

Where each requirement came from: PR = project request, MT = coordination meeting, EM = photo email thread, PL = data protection policy, LM = logistics memo, FB = field feedback messages, FM = paper needs form.

ID	Requirement	Source
FR-01	A field coordinator can fill in a needs form on a phone: location, number of families and people, how many are children under 5, elderly or disabled, what items are needed (food parcels, water, blankets, medical kits, tents, hygiene kits) and how many, urgency from 1 to 3, and access conditions. It works offline and sends when the internet is back.	PR, FM, MT
FR-02	The Coalition Planner approves or rejects the form and gives a reason. Approved forms become open requests that all charities can see, showing how much is still needed and what is already promised or on the way.	PR, MT
FR-03	The system stores donors (person, company, international organization). A donor can pledge money, or goods, but goods only for an open request. If the pledge is more than what is needed, the system warns them.	PR
FR-04	If a company pledges more than 1,000 items it has to book a delivery slot at a warehouse. The donor gets a confirmation, and we keep track of booking requests nobody answered.	PR, FB
FR-05	Warehouse staff check the arriving goods against the pledge and write down what was accepted and what was rejected. A rejected item (expired or damaged) needs a reason.	PR
FR-06	Accepted goods are put in a batch with an expiry date. The system prints batch labels in Arabic and English.	PR, FB, LM
FR-07	The system keeps stock per warehouse (Cairo, Ismailia, Arish) and per batch, including expiry date and whether it must stay cool. It suggests the batch that expires first and warns when a batch is close to expiring.	PR, LM, MT

ID	Requirement	Source
FR-08	The Logistics Officer plans a convoy from a warehouse across the border to the distribution points. The system checks each truck's weight and size limit, and only lets items that must stay cool (like medicines) go on cooled trucks.	PR
FR-09	For every truck the system stores capacity, cooling, driver, and permit number and status. It also stores the border's daily quota and produces the cargo list needed for each permit.	PR, LM
FR-10	When a permit is rejected or the quota is cut, the system helps reschedule the convoy so the most urgent needs go first. It also shows which perishable items are at risk.	PR, MT
FR-11	Each family is stored with family ID, size, location, vulnerability category and QR card number. We only collect what is needed to deliver aid.	PR, PL
FR-12	Staff can find a family using the QR code, or with the family ID or a masked national ID search if the card is damaged.	FB
FR-13	The system blocks giving the same category to the same family within 14 days, across all charities. Other charities only see that it is blocked, not the family's details.	PR, PL, FB
FR-14	Every handover is saved: family, category, quantity, staff member, date and whether it was done offline. Handovers work offline for a full day and sync later.	PR, LM
FR-15	Staff can flag elderly or disabled families so they are served first.	FB
FR-16	Money donations turn into purchase orders (supplier, items, amount). A purchase over 500,000 EGP needs Finance Committee approval, and the goods bought stay linked to the donor.	PR
FR-17	For each donation the system can create a PDF report in Arabic and English showing where it went, with at least the distribution point and date.	PR, MT, LM
FR-18	Proof of delivery is photos of boxes and distribution sites, plus handover counts with time and place. We don't need photos of families. A photo of a person is only saved with their consent, and never a child's.	EM
FR-19	The system can answer four questions: which needs are still not covered (by location and category), how long items take from donation to handover, how many families got each category this month, and which donors' items reached which locations.	PR
FR-20	What a user sees depends on their role. A charity only sees its own families, and only approved staff can see names and photos. Each charity gets 2 to 10 staff accounts, volunteers can use phones, and the whole app works in Arabic and English.	PR, PL, LM
FR-21	National ID numbers are stored encrypted and shown masked. Personal data is deleted 2 years after the last distribution, and only summary numbers stay.	PL
FR-22	Family data can only be shared outside the Coalition after the board approves it. The system keeps a log of who viewed or changed personal data.	PL, PR

Optional: if our team has more than three people, we might add a volunteer driver program (license checks and trip assignments) and a simple forecast of needs based on data from similar past crises.

4. Requirements Elicitation Techniques

We need a mix of techniques because our stakeholders are so different. Some senior people know the official rules and set priorities, while field staff in the 14 charities know what actually happens. Other groups, like the families and the border authority, are tough to reach.

Technique	Who we use it with	Why we picked it
Reading documents	Needs form, protection policy, logistics memo, Excel files, UN cluster and Sphere standards	It is quick and cheap, and the documents contain rules we can't get wrong, like the 14-day rule and the 500,000 EGP limit. It also covers the research questions in the request.
Interviews	Heba, Hesham, Rasha, Dr. Tamer, Ahmed, the Logistics Officer and the Planner	Only a few people take the big decisions, so a conversation where we can ask follow-up questions works best.
Questionnaire	Field, warehouse and distribution staff and volunteers	There are a lot of users, they are busy and often offline. A short form in Arabic and English reaches more of them than interviews would.
Watching people work	A field needs assessment, warehouse receiving, a distribution line	People skip steps when they describe their job. Watching shows the delays and workarounds, like long lines and wet QR cards.
Workshop	Rasha, Dr. Tamer, Heba and our team	Donor proof and family privacy pull in different directions. We would rather get both sides in one room than guess.
Prototypes	Field, warehouse and distribution staff	Rough screens get users reacting early and show us what we missed.

We're skipping a big brainstorming session because the problem is already pretty clear. We also won't interview families directly, to protect their privacy. Instead we'll rely on what field staff tell us and what we can observe.

5. Interview Agendas

5.1 Internal operational stakeholder: Ahmed Badawi, Warehouse Manager (Arish)

Time: 45 minutes (in Arish or on a video call). **Goal:** find out how goods are currently received, checked, labeled, stored and sent out, and what problems come up. **Agenda:** intro (5 min), receiving and checking goods (10), batches and stock (10), delivery slots and convoy delays (10), what he needs from the system (7), wrap up (3).

1. Can you walk us through a typical day in your warehouse?
2. What happens between when a donation truck arrives and when the goods are placed on shelves? How do you check them against the pledge?
3. Why do you usually reject items, and how do you tell the donor?
4. How are batch labels made right now? Who actually reads them?
5. How do you pick which batch goes out first? Can you explain what went wrong with the food that expired last year?
6. How do you handle items that need cooling, and how much cold storage do the three warehouses have?
7. How do you track the stock in Cairo, Ismailia and Arish, and how do you know you can trust those numbers?
8. How do you arrange delivery slots for big donations? What do you need to know if a convoy is delayed or a permit gets rejected?
9. If you could only fix one thing, what would it be? And what documents can you share with us?

5.2 Internal executive stakeholder: Heba Elmasry, Coalition Logistics Director (sponsor)

Time: 60 minutes (Coalition office in Cairo). **Goal:** get clarity on the overall goals, priorities, decision rules, limits and risks for the whole Coalition. **Agenda:** intro (5 min), goals and priorities (15), decision rules and approvals (15), donors, privacy and limits (15), risks and optional features (7), wrap up (3).

1. What would make the first version a real success after four months? What about after a year?

2. Which problems last year were the most damaging: lost aid, lost money or lost donor trust?
3. How do the 14 charities share information now, and do they actually want one shared system?
4. If we have to keep things simple at first, what features can't we skip?
5. Who can approve needs forms, change how pledges are matched, and add charities or users?
6. When a quota is cut or permits get denied, who decides which needs go first, and how?
7. What should donor reports include? How do you want to handle the donor photo issue?
8. Are there any laws, board rules or border regulations on what data we can save, share or host? Who approves purchases over 500,000 EGP, and how quickly?
9. What worries you the most, and how will charities be trained to move away from Excel?

6. Questionnaire for Operational Users

Who it's for: field coordinators, warehouse staff, distribution staff and volunteers. **Format:** about 10 minutes, in Arabic and English, easy to fill in on a phone, anonymous.

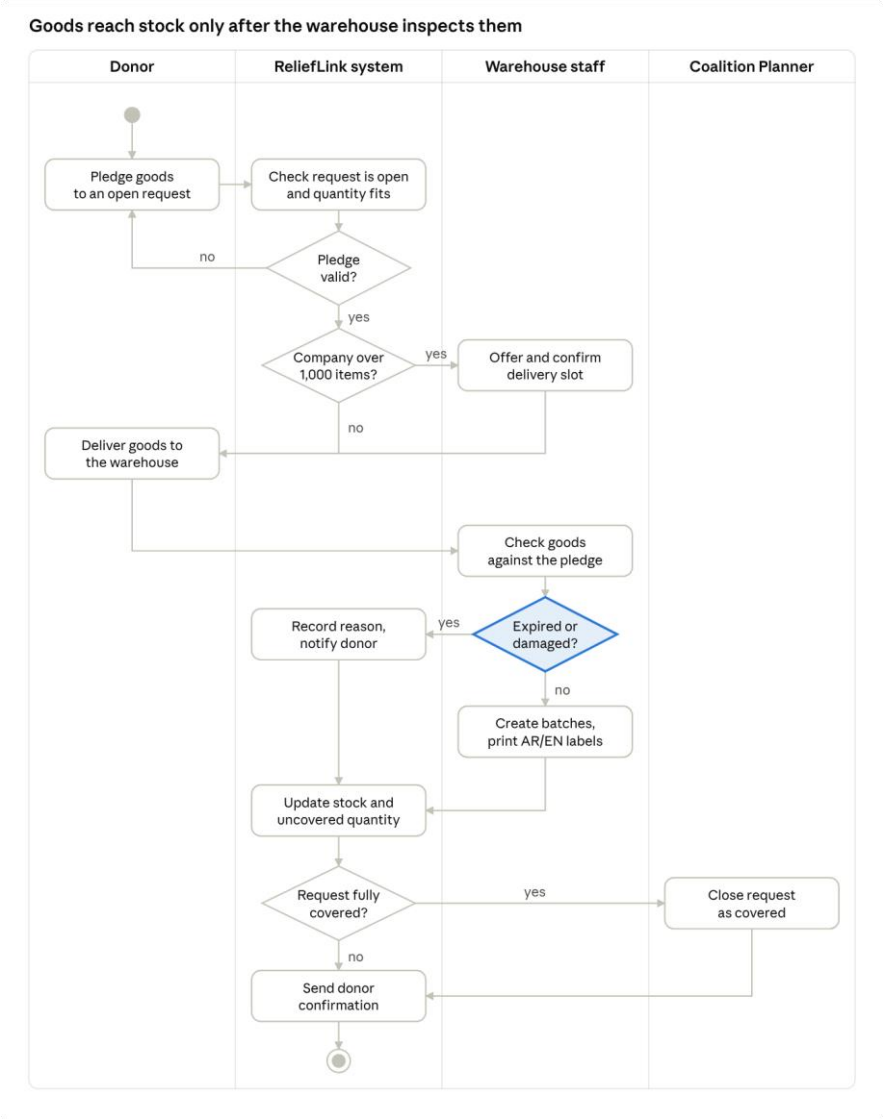
Intro for people filling it in: Hi! We're a student team building a system to help the Coalition match needs with donations, so aid gets to families more fairly. Your answers help us design it around the way you work. There are no right or wrong answers.

1. What's your main job? (Field coordinator / Warehouse staff / Distribution staff / Volunteer / Other)
2. What do you use for work? (My own phone / A shared phone / Computer / Paper only)
3. How often do you have internet at work? (Always / Sometimes / Rarely / Never)
4. How are things recorded now? (Paper / Excel / WhatsApp / Other)
5. What language do you prefer for screens and labels? (Arabic / English / Both)
6. About how long does it take to fill in a needs form? (under 30 min / 30 to 60 min / 1 to 2 hours / over 2 hours)
7. How often are needs forms lost, hard to read or sent late? (Never / Sometimes / Often)
8. How often do items arrive expired or damaged? (Never / Sometimes / Often)
9. How easy is it to know which batch to send out first? (1 = very hard, 5 = very easy)
10. How long does it usually take to serve one family? (under 1 min / 1 to 3 min / 3 to 5 min / over 5 min)
11. How often do QR cards stop working? (Never / Sometimes / Often) What do you do when that happens?
12. How often did you serve a family that already got the same item from another charity? (Never / Once / Many times)
13. How important are the following? (1 = not important, 5 = very important): working offline all day, scanning QR codes, a warning about the 14-day rule, a tip on which batch goes first, Arabic labels and screens.
14. What's one thing that would save you the most time?
15. Is there anything about privacy or family safety we should keep in mind?

7. Workflow Documentation

We picked two of the more complicated business processes and modeled them as activity diagrams with swimlanes.

7.1 Process 1: Goods pledge, delivery slot and warehouse acceptance



7.2 Process 2: Convoy planning, permit and rescheduling

A rejected permit or cut quota sends the convoy back to rescheduling

